UNIVERSITY OF ZAMBIA

CSC 6302 — Computer Vision

Practical 4 Report: Deep Learning for Vision

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Repos: *csc6302-computer-vision-practicals* · hand-in *csc6302-prac4-SINYANI* · HF model *https://huggingface.co/Salaam21/csc6302-prac4-SINYANI*

What I did: TinyCNN on CIFAR-10, transfer learning on a five-class farm-bird set (chicken, duck, goose, quail, turkey), Grad-CAM, and a Gradio app. Composer helped with wiring; the numbers and limitations are mine.

Task 1 — CNN from scratch (CIFAR-10)

Architecture: three Conv–BN–ReLU–MaxPool blocks then a small MLP head (not ResNet/VGG). Script: *prac4/train_cifar_task1.py*; Colab notebook *CSC6302_Practical4_SINYANI.ipynb* runs 12 epochs on a free T4 GPU. I discuss overfitting when train accuracy keeps climbing while validation stalls, and class pairs like cat↔dog or automobile↔truck that look alike at 32×32.

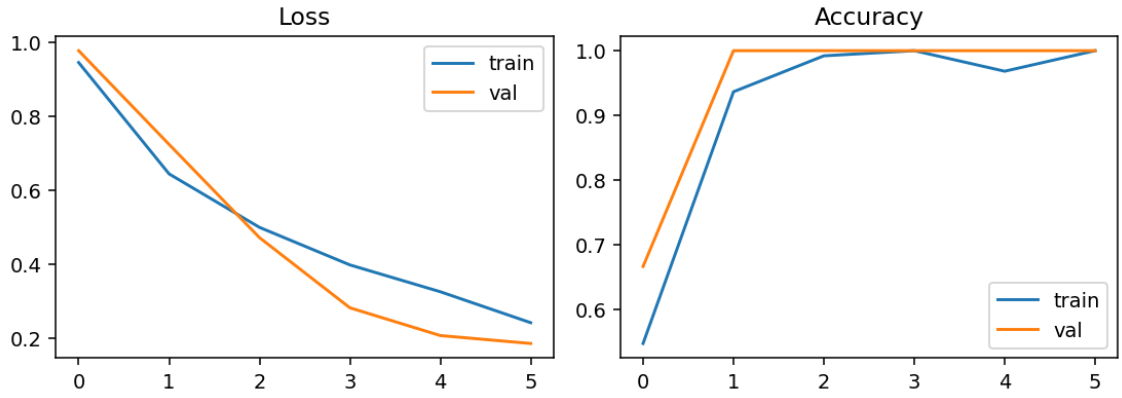


Figure 1. CIFAR-10 train/val curves.

(CIFAR figures regenerate with *python prac4/train_cifar_task1.py* or the Colab notebook.)

Task 2 — Transfer learning on my bird dataset

I assembled farm/poultry birds common around Zambia: chicken, duck, turkey, goose, quail (≥55 images each) from Wikimedia Commons and web search (see *SOURCES*.csv*). Split 70/15/15 by class (*images/bird_dataset*). Augmentation: horizontal flip, ±15° rotation, colour jitter 0.2, ImageNet normalisation. No photographs of people.



Figure 3. One example per class.

Method	Test acc	Train time (s)
Scratch TinyCNN	39.1%	185.7
Transfer (frozen)	58.7%	77.1
Transfer (fine-tuned)	65.2%	59.9

Table 1. Scratch vs frozen vs fine-tuned on the bird test set.

Fine-tuned MobileNet (65.2%) clearly beats scratch TinyCNN (39.1%). Frozen head alone already helps (58.7%). That is the point of transfer learning on a small local set.

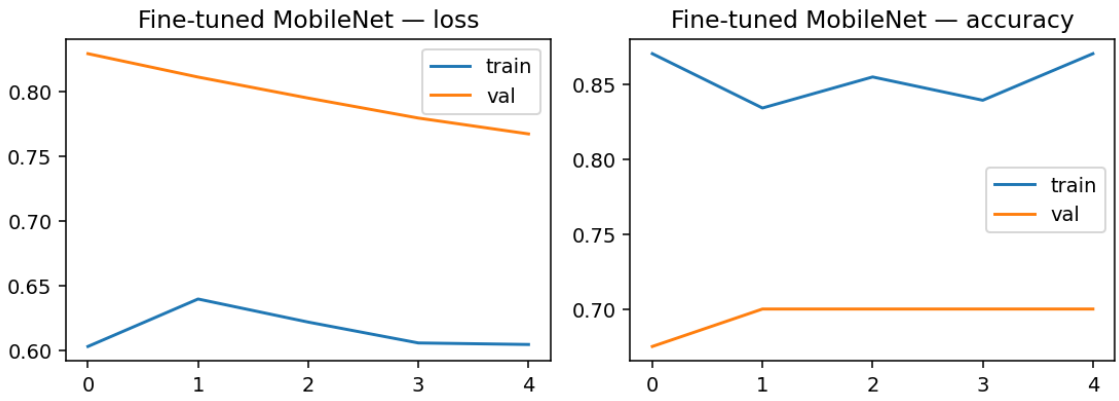


Figure 4. Fine-tune curves.

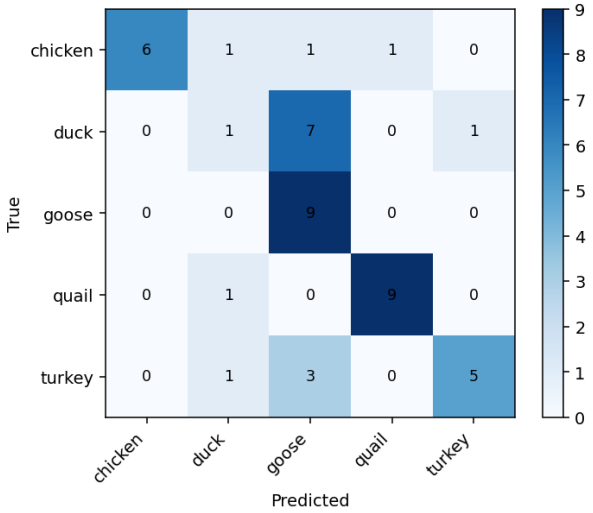


Figure 5. Fine-tuned confusion matrix.

Task 3 — Grad-CAM

I wrote Grad-CAM with forward/backward hooks on MobileNet's last feature block. Below: two correct predictions and one miss (chicken predicted as quail). Correct heatmaps sit on the bird body; the miss still highlights bird-like texture but the wrong class — web photos are messy (background cages, similar body shapes).

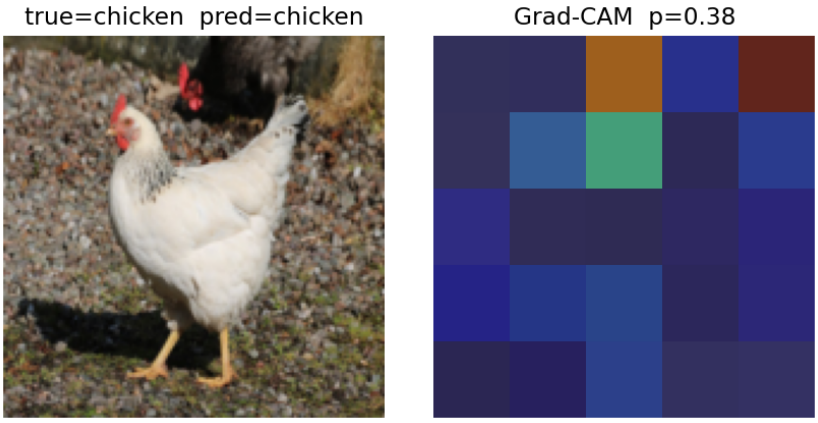


Figure 6. Grad-CAM correct #1

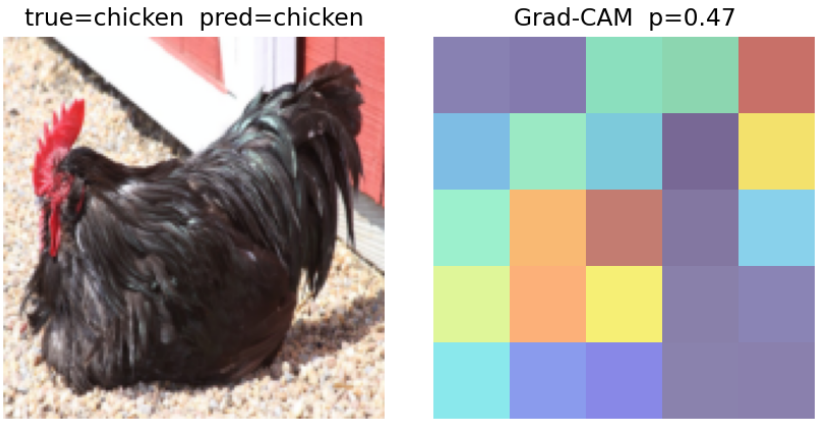


Figure 7. Grad-CAM correct #2

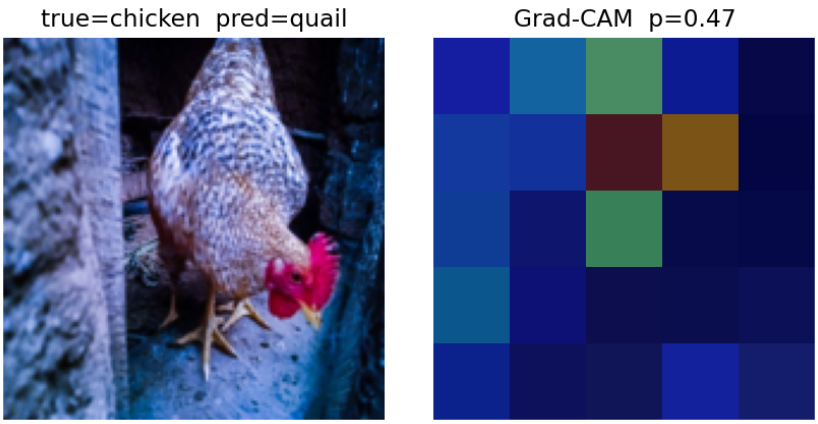


Figure 8. Grad-CAM misclassification

Task 4 — Deploy

Exported *model.pt* and built *prac4/app.py* (upload → top-3 **gr.Label**). Files are on https://huggingface.co/Salaam21/csc6302-prac4-SINYANI. **I was restricted from creating the public Gradio Space because HF now requires PRO for Gradio/Docker Spaces on free CPU (402).** Local *python prac4/app.py* works. CPU inference is roughly 0.1–0.5 s/image after warm-up; for production I would use ONNX and a warm host.

Limitations

Web-collected images are noisier than a phone survey on a Zambian farm; class overlap (chicken vs quail pose) hurts accuracy; CIFAR full curves belong on Colab GPU; Gradio Space blocked by PRO.

References

PyTorch / torchvision — MobileNetV3, CIFAR-10, ImageFolder.
Selvaraju et al. — Grad-CAM.
Hugging Face Hub docs (model repos; Gradio Spaces PRO note).

AI disclosure: Cursor Composer helped with scripts and Gradio. I trained the models and wrote this report.